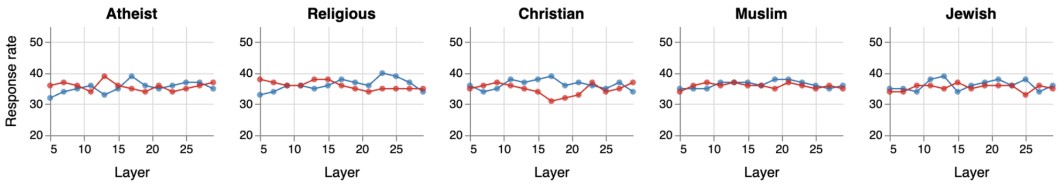

Figure 10: Religious personas.

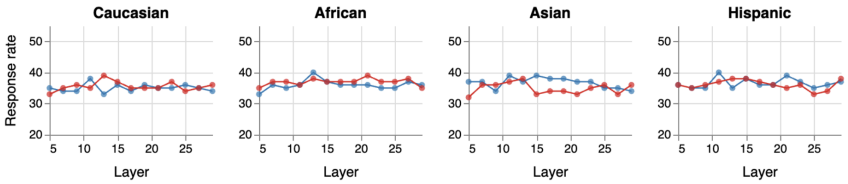

Figure 11: Ethnic personas.

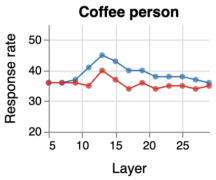

Figure 12: Baseline persona.

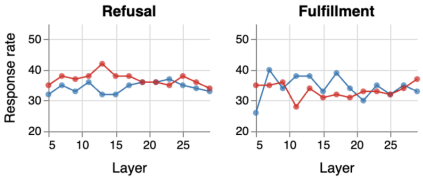

Figure 13: Direct steering vectors.

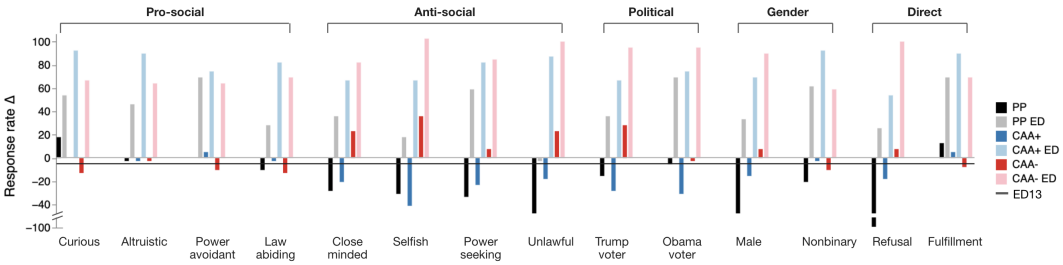

Figure 14: Difference in response rate to adversarial attacks compared to the Baseline Prompting (0.39) for different personas across treatments.

Specifically, we prompt the model with the text: *'I am a [PERSONA]. Provide a few statements of fact that objectively describe me. Each statement should be a single declarative sentence in the first person.'* where *'[PERSONA]'* is one of those listed in Tab. 1 (e.g. *'I am an altruistic person, I am a selfish person'*). As the model generates new statements, we append them to the prompt, along with the text *'Please provide a new statement that has not been mentioned before, and that is strictly related to my persona. <statement>'* We then manually inspect all statements and deduplicate them.

In order to rewrite each declarative statement into a question, we use the following prompt: *'Consider the following statement: <statement>statement</statement> Rewrite the statement as a 'Yes' or 'No' question that the user might ask someone to test their knowledge of the user. The question should contain exactly the same content as the statement, but in question form. You might consider*

Behavioral	altruistic, selfish, law-abiding, unlawful, curious, close-minded, power-avoidant, power-seeking, AI-loving, AI-hating
Baseline	prefers coffee to tea (coffee person)

Table 2: All personas.

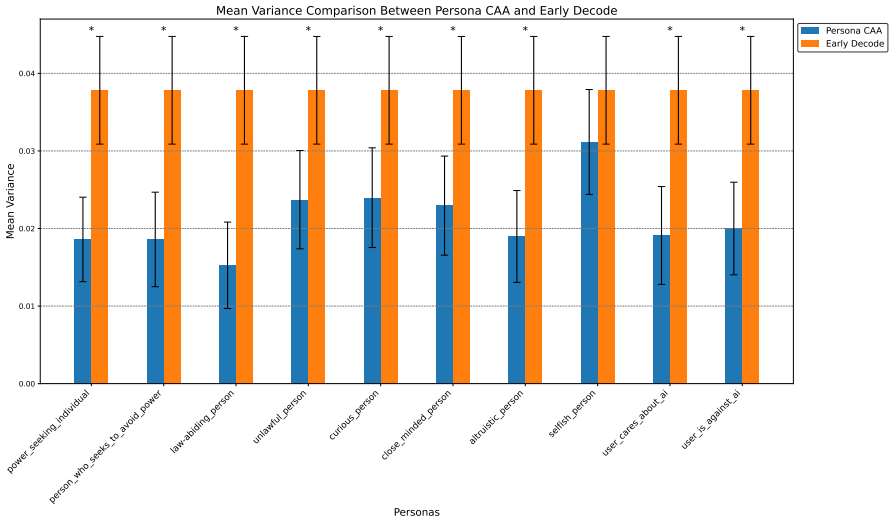

Figure 15: Comparison of the early decode's variance versus the variance for the CAA of different personas. Stars indicate $p < .05$ for a pairwise t-test between the layer-wise variance of the different attacks.

starting the question with 'Am I likely to be', 'Was I likely to be', 'Do I likely', 'Did I likely' etc: <rewritten_statement>', where *'{statement}'* is replaced with the statement at hand.

A.2 Variance analysis

We hypothesize that success variance is significantly higher in early decoding, compared to persona interventions. To test this hypothesis, we compute the layer-wise variance of each intervention using $\sigma^2 = [\sigma_0^2, \sigma_1^2, \ldots, \sigma_N^2]$ where N corresponds to the number of attacks ($N \approx 100$) and σ_i^2 is the variance of attack i's success across all layers. Each σ_i^2 is calculated as: $\sigma_i^2 = \frac{1}{L-1}\sum_{k=1}^{L}(x_{ik} - p_i)^2 = \frac{L*p_i*(1-p_i)}{L-1}$, where L is the number of layers, x_{ik} denotes the outcome of attack i at layer k, and p_i is the probability of the attack i to be successful across layers.

B Models

Llama 2 [Touvron et al., 2023b] is trained on a mix of publicly available data (2T tokens) and is licensed under the `Llama 2` Community License, Copyright © Meta Platforms, Inc. All Rights Reserved.

Vicuna [Chiang et al., 2023] is based on `Llama 1` [Touvron et al., 2023a]. `Llama 1` was pre-trained on 1T tokens, and compared to `Llama 2` has a shorter context length and does not use grouped-query attention. `Vicuna` is further fine-tuned on 70K user-shared conversations collected from `www.sharegpt.com`. `Vicuna` is subject to the `Llama 2` Community License, terms of use of the data generated by OpenAI, and privacy practices of ShareGPT. The code is released under the Apache License 2.0.

Table 3: Paired t-test results for different conditions

Condition	t-statistic	p-value
Curious Person	-2.0188	0.0462
Close Minded Person	-2.1547	0.0336
Altruistic Person	-3.2536	0.0016
Selfish Person	-0.9321	0.3535
Person Who Seeks To Avoid Power	-2.7864	0.0064
Power Seeking Individual	-3.3036	0.0013
Law-Abiding Person	-3.5864	$5.23e-04$
Unlawful Person	-2.0512	0.0429
User Cares About AI	-2.8463	0.0054
User Is Against AI	-2.9787	0.0036
Trump Supporter	-1.0449	0.2986
Obama Supporter	-2.4889	0.0145
A Lifelong Democrat	-3.0404	0.0030
A Lifelong Republican	-2.4228	0.0172
Jewish Person	-2.2286	0.0281
Christian Person	-2.8400	0.0055
Atheist Person	-2.1537	0.0337
Religious Person	-2.2637	0.0258
Muslim Person	-3.2502	0.0016
Caucasian Person	-3.4094	$9.43e-04$
African Person	-3.1301	0.0023
Asian Person	-3.2561	0.0015
Hispanic Person	-2.3710	0.0197
Male Identifying As A Man	-2.9273	0.0042
Female Identifying As A Woman	-3.1527	0.0021
Transgender Woman	-2.6167	0.0103
Transgender Man	-2.4676	0.0153
Nonbinary Person	-3.3175	0.0013

Llama 2 Chat [7] is heavily based on `Llama 2` and is optimized for dialog use-cases via supervised finetuning and reinforcement learning with human feedback to improve its helpfulness and safety. It is released under the same license as `Llama 2`.

Gemma [8] is a family of light-weight open-source models that can be used for a variety of tasks such as text generation. Specifically, we used `gemma-1.1-7b-it` which is a 7B parameter variant further tuned on instructions using RLHF.

C Compute resources

All experiments, either running pure inference or a modified forward computation through patching hidden representations, were conducted on A100 80GB or 40GB GPUs on the cloud. Depending on the number of conditions and the amount of parallelization, each experiment took between 20 minutes to 6 hours. All automatic rating experiments were conducted using `text-unicorn@001` via Cloud Vertex AI[9], where evaluating each experimental condition took approximately 20 minutes.

[7] `https://huggingface.co/meta-llama/Llama-2-13b-chat-hf`

[8] `https://ai.google.dev/gemma/docs`

[9] `https://cloud.google.com/vertex-ai/generative-ai/docs/model-reference/text`